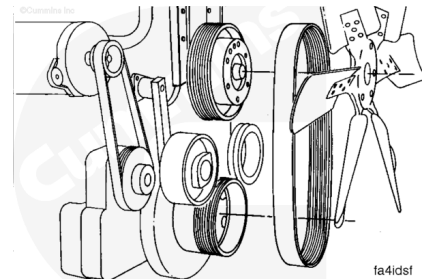


Trainings ▾

Preparatory Steps

- Remove the fan. Refer to Procedure 008-040 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-040-tr.html)
- Remove the fan belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html)

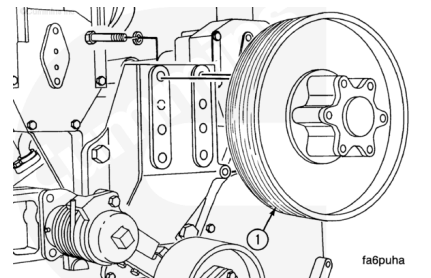


Remove

Standard

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



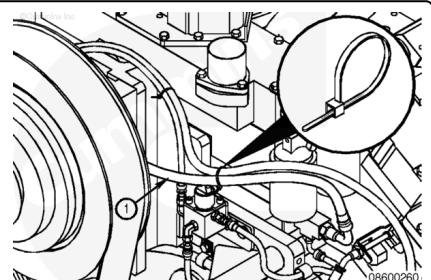
Remove the eight capscrews and the fan hub (1).

Rockford

Remove the wire tie that secures the number 12 drain hose (1) to the number 8 supply hose.

Disconnect the drain hose from the port in the side of the fan clutch.

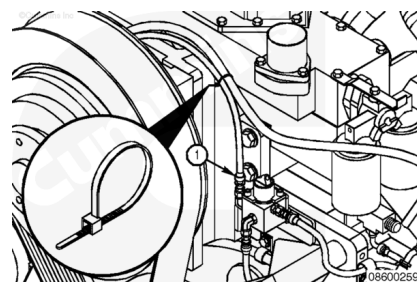
Disconnect the drain hose from the large tee in the front gear housing and remove the hose.



Remove the wire tie that secures the number 4 solenoid-to-clutch supply hose (1) to the number 8 fan hub supply hose.

Disconnect the solenoid-to-clutch supply hose from the top of the fan clutch.

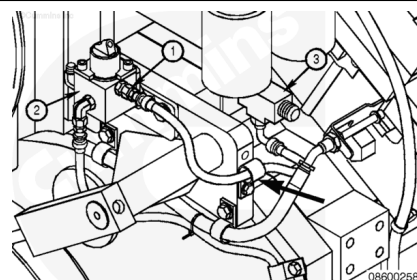
Disconnect the solenoid-to-clutch supply hose from the solenoid and remove the hose.



Remove the capscrew and p-clip located on the side of the actuator.

Disconnect the solenoid drain hose from the sump port (1) of the solenoid (2).

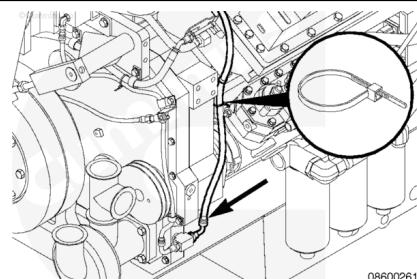
Disconnect the solenoid drain hose from the lower port of the large tee (3) and remove the hose.



Remove the wire tie that secures the number 8 fan hub supply hose.

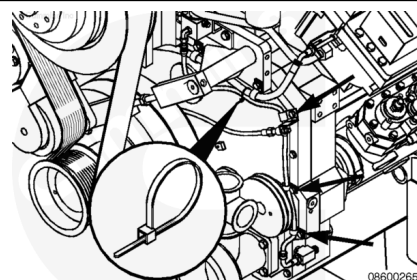
Disconnect the fan clutch supply hose from the rear side of the lower tee.

Disconnect the fan clutch supply hose from the top of the fan clutch and remove the hose.



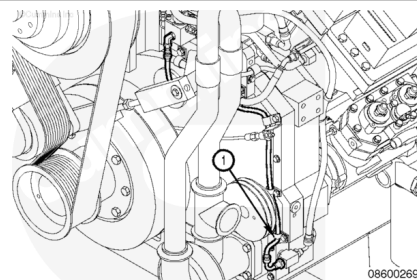
Remove the wire tie that secures the solenoid supply to the fuel hose.

Remove the three capscrews and p-clips that secure the solenoid supply hose to the front of the engine.

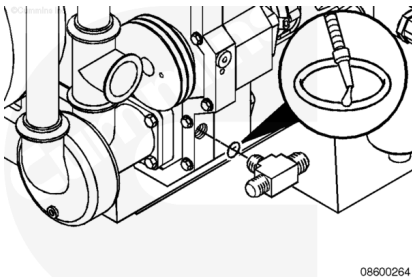


Disconnect the solenoid supply hose from the supply port of the solenoid valve.

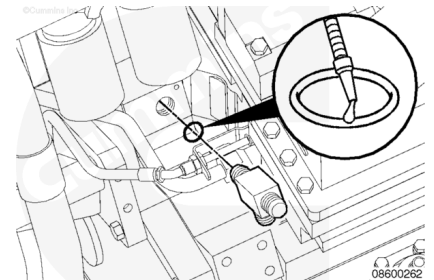
Disconnect the solenoid supply hose from the lower tee and remove the hose.



Remove the tee located in the lower left bank port in the front gear cover.

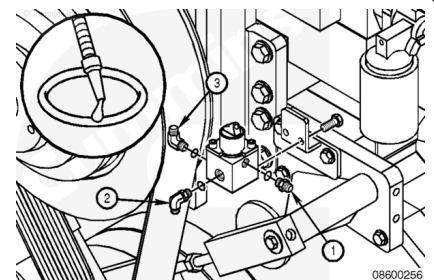


Remove the tee located in the top left bank of the front gear cover.



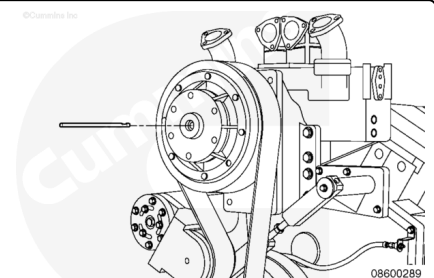
Remove the following:

- Sump port male union (1)
- Supply port 90 degree elbow (2)
- Control port 90 degree elbow (3)
- Solenoid.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



For fan clutches equipped with speed sensors, remove the speed sensor from the center of the fan clutch hub.

Remove the capscrews and the fan hub.

Inspect for Reuse

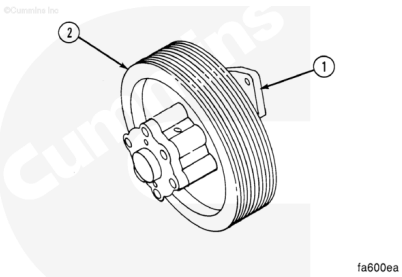
Standard

Check the seal for leaks.

Check the shaft (1) for damage.

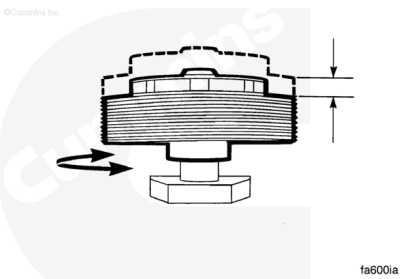
Check the fan hub exterior (2) for damage.

If any of these components are damaged, the fan hub **must** be rebuilt.



Check the fan hub end clearance.

Bearing End Clearance		
mm		in
0.025	MIN	0.001
0.25	MAX	0.010



If the clearance is **not** within the specifications, the fan hub **must** be rebuilt.

Rotate the pulley to check for rough or damaged bearings.

If the fan hub does **not** rotate freely, the fan hub **must** be rebuilt.

If applicable, remove the pipe plugs (3). Install a grease fitting in one of the holes.

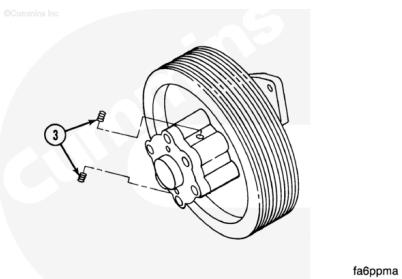
Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or its equivalent.

If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

Use a grease gun and pump grease into the hub until it begins to come out of the open pipe plug hole.

Remove the grease fitting.

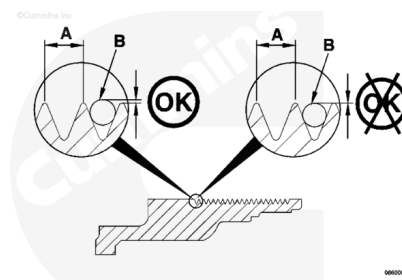
Install the pipe plugs (3).



Torque Value: 15 n•m [133 in-lb]

Poly Vee pulley groove wear can be measured by placing a gauge ball or drill bit in the pulley groove and checking the protrusion with a straight edge. If the gauge ball or drill bit is below the pulley peaks, the pulley **must** be replaced.

All QSK45 and QSK60 engines use the L section pulley belt type.



Poly Vee - Pulley Reuse Guidelines

Groove peak to peak dimension

Measuring tool dimension (B)

4.70 mm [0.185 in]

4 mm [0.16 in] gauge ball or a number 20 drill bit

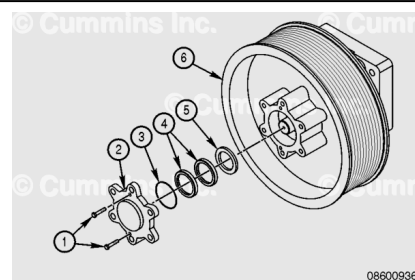
Disassemble

Standard

Remove the following parts from the pulley:

1. Capscrews
2. Fan pilot
3. O-ring
4. 2 locknuts
5. Plain hardened washer.

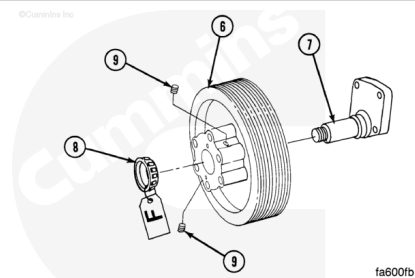
Discard the o-ring.



Note : If force is required, use an arbor press. Support the pulley (6) and push on the small end of the shaft (7).

Remove the pulley assembly (6) from the shaft (7).

Remove the front roller bearing race (8).



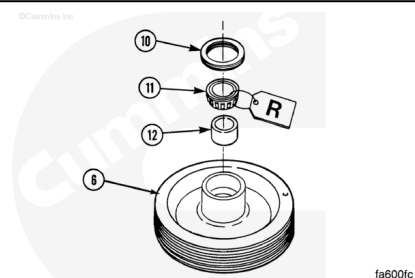
Turn the pulley assembly (6) over.

Remove the oil seal (10).

Remove the rear roller bearing race (11).

Remove the inner bearing spacer (12).

Use a mandrel to remove the front end rear bearing cones.

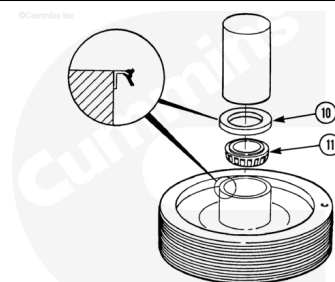


Assemble

Standard

⚠ CAUTION ⚠

Do not lubricate the seal surface of the shaft. The seal and seal surface must be clean and dry.



Lay the fan hub down on the side with the tapped holes in the cover.

Use a mandrel to install the first bearing outer. Press it down until it touches the shoulder.

Grease the bearing (11).

Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or its equivalent.

If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

Install the bearing.

Use a mandrel to install the grease seal (10). Press it down so that it is flush with the outer face of the fan hub.

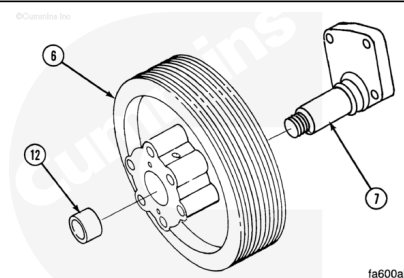
Install the seal with the flat surface uppermost.

Note : Excessive grease will affect the end float reading.

Turn the fan hub over.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use a mandrel to install the second bearing outer. Press it until it touches the shoulder.

The rear bearing **must** slide easily on the shaft. If it does **not**, check for chips or burrs.

Install the inner bearing spacer (12).

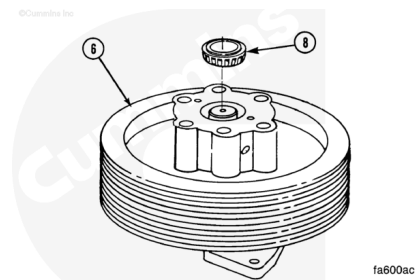
Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or its equivalent.

If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

Fill the cavity with 360 cc [12 oz] of grease.

Grease the bearing roller race with grease. Install into the bearing outer just fitted.

Remove any excess grease.

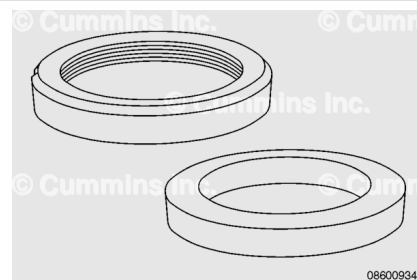


Install the plain spacer washer.

Install the first nut with the chamfer facing away from the fan hub.

Torque Value: 600 n•m [443 ft-lb]

Continually rotate the pulley wheel while applying the torque.



Apply Loctite™ 648, or equivalent, to the second locknut.

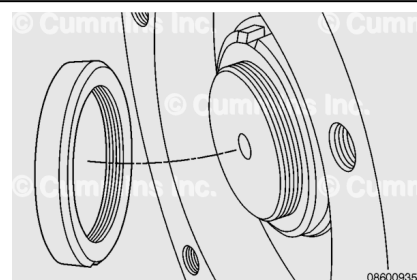
Install the second locknut with the chamfer facing toward the fan hub.

Torque Value: 250 n•m [184 ft-lb]

Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or its equivalent.

If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

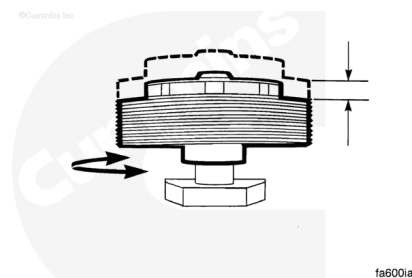
Fill the double nut area with 300 cc [10 oz] of grease.



Use a dial indicator to measure the bearing end clearance.

Bearing End Clearance

mm		in
0.025	MIN	0.001
0.105	MAX	0.004



If the clearance is **not** within the specifications, rebuild the fan hub with a suitable size spacer.

Check the bearing end clearance.

⚠ CAUTION ⚠

Do not use more grease than specified. Too much grease will cause excessive heating and failure.

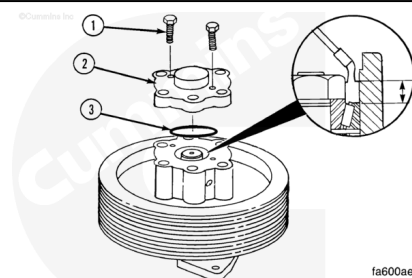
Install the o-ring (3) on the fan pilot (2).

Lubricate the o-ring with vegetable oil.

Install the fan pilot and o-ring.

Install the capscrews (1).

Torque Value: 20 n·m [177 ft-lb]



Install

Standard

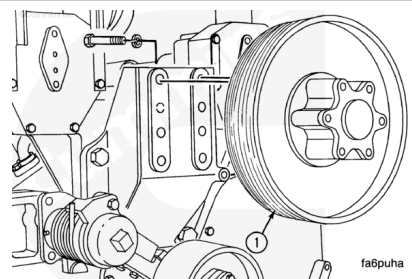
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Install the fan hub, the eight washers, and the capscrews.

Tighten the fan hub capscrews.

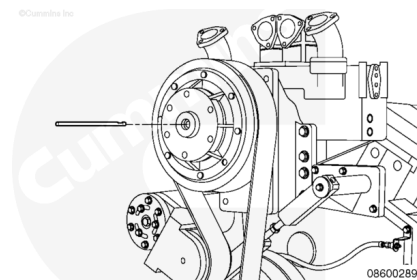
Torque Value: 280 n·m [207 ft-lb]



Rockford

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the fan hub and capscrews.

Tighten the capscrews.

Torque Value: 280 n•m [207 ft-lb]

For fan clutches equipped with speed sensors, install the speed sensor and tighten.

Torque Value: 30 n•m [22 ft-lb]

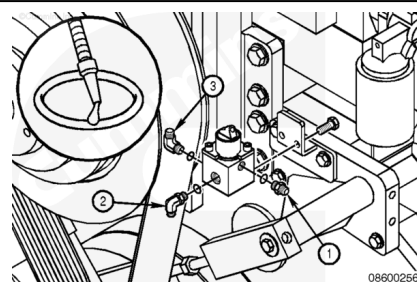
Install the solenoid onto the solenoid bracket with capscrews and tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]

Install the o-rings onto the three male connections and lubricate with clean engine oil.

Install the following and tighten:

- Straight male union into sump port (1).
- Elbow into supply port (2)
- Elbow into control port (3).



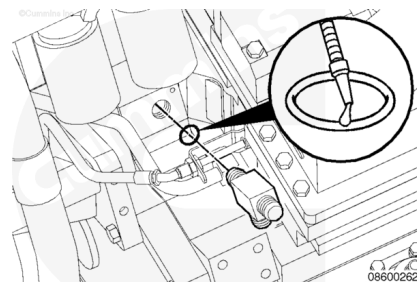
Torque Value: 24 n•m [212 in-lb]

Install the o-ring onto the tee and lubricate with clean engine oil.

Install the tee into the top left bank of the gear cover and tighten the tee.

Torque Value: 47 n•m [35 ft-lb]

The final position of the center hose port is facing down.

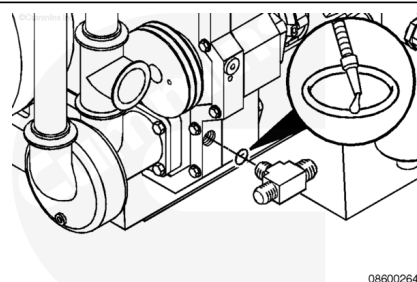


Install the o-ring onto the tee and lubricate with clean engine oil.

Install the tee into the lower left bank port in the front gear cover and tighten the tee.

Torque Value: 11 n•m [97 in-lb]

The final position of the tee is with the number 4 hose connection facing forward and the number 8 hose connection facing to the rear.

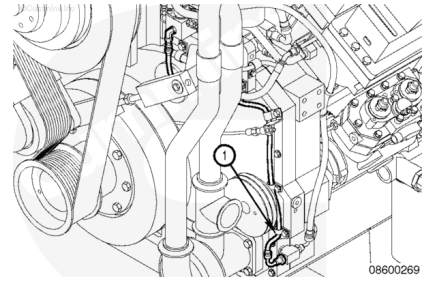


Connect the number 4 solenoid supply hose to the supply port of the solenoid valve.

Route the supply hose to the lower tee in the front gear cover and connect it to the tee.

Tighten the hose nuts.

Torque Value: 48 n•m [35 ft-lb]



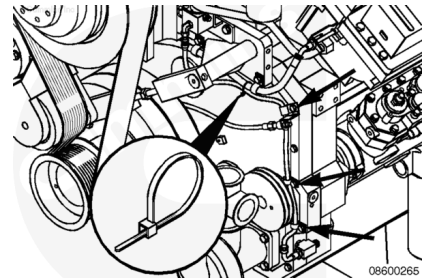
Attach the hose with two p-clips to the lower front gear cover capscrews located behind the accessory drive pulley.

Install a third P-clip under the front trunnion remote grease nipple bracket.

Secure the solenoid supply hose to the fuel hose using wire ties.

Tighten the P-clip capscrews.

Torque Value: 80 n•m [59 ft-lb]



The lower hose elbow faces slightly outboard, and the top elbow faces slightly to the rear to allow the number 4 hose elbow to clear the number 8 supply hose.

Connect the number 8 fan hub supply hose to the top port of the fan clutch.

Route the hose to the lower tee on the front gear cover and connect the hose.

Tighten the hose nuts.

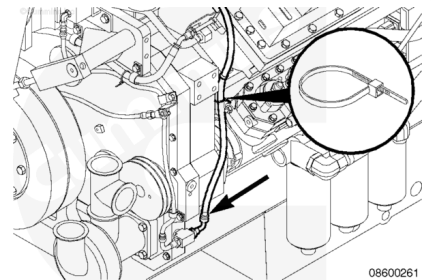
Torque Value: 48 n•m [35 ft-lb]

For engines equipped with an air compressor, secure the number 8 fan hub supply hose to the water tube with a wire tie.

For engines that are **not** equipped with an air compressor, secure the fan hub supply hose at a higher location with a wire tie.

Secure the hose with a P-clip installed to the front lower air intake manifold capscrew.

Torque Value: 16 n•m [142 in-lb]



Connect the number 4 solenoid drain hose to the straight male connector in the sump port (1) of the solenoid (2).

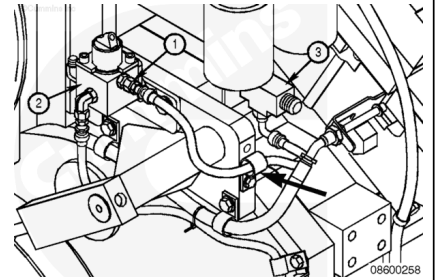
Route the drain hose to the lower port in the upper front gear cover tee (3) and connect the hose.

Tighten the hose nuts.

Torque Value: 16 n•m [142 in-lb]

Secure the hose with a P-clip connected to the actuator bracket.

Torque Value: 16 n•m [142 in-lb]



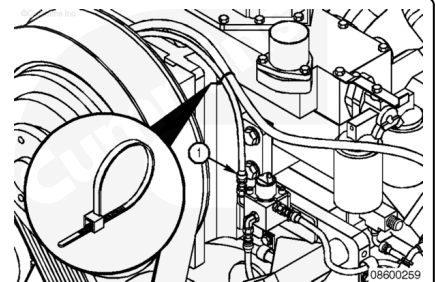
Connect the number 4 solenoid-to-clutch drain hose to the control port (1) in the solenoid.

Route the hose to the top of the fan clutch and connect the hose.

Tighten the hose nuts.

Torque Value: 16 n•m [142 in-lb]

Secure the hose to the number 8 fan supply hose with a wire tie.



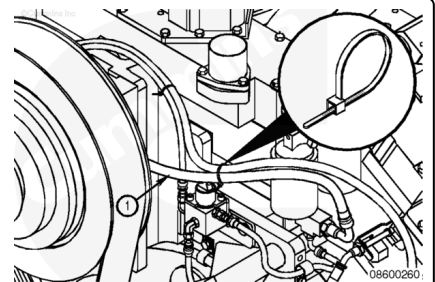
Connect the number 12 drain hose (1) to the connector in the side of the fan clutch.

Route the hose to the upper tee in the front gear cover and connect the hose.

Tighten the hose nuts.

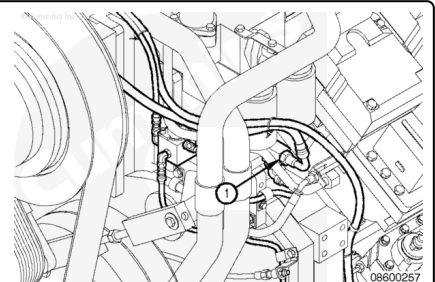
Torque Value: 95 n•m [70 ft-lb]

Secure the hose to the fan hub supply hose with wire ties.



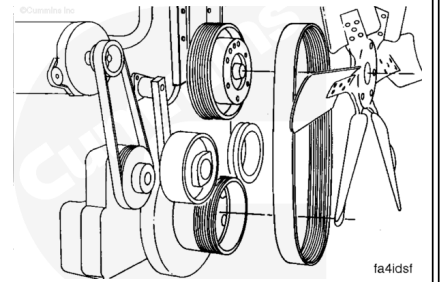
Secure the three hoses connected to the fan clutch together to provide an orderly routing.

Make sure all of the hoses are routed so they are clear of any rotating parts.



Finishing Steps

- Install the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html>)
- Install the fan. Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-040-tr.html>)
- Operate the engine and check for proper operation.



Last Modified: 07-May-2015
